1.4 Scope and Applications

In ancient times the scope of statistics was limited. When people hear the word 'Statistics' they think immediately of either sports related numbers or a subject they have studied at college and passed with minimum marks. While statistics can be thought in these terms there is a wide scope for statistics. Today, there is no human activity which does not use statistics. There are two major divisions of statistical methods called descriptive statistics and inferential statistics and each of the divisions are important and satisfies different objectives. The descriptive statistics is used to consolidate a large amount of information. For example, measures of central tendency, like mean are descriptive statistics. Descriptive statistics just describes the data in a condensed form for solving some limited problems. They do not involve beyond the data at hand.

Inferential statistics, on the other hand, are used when we want to draw meaningful conclusions based on sample data drawn from a large population. For example, one might want to test whether a recently developed drug is more efficient than the conventional drug. Hence, it is impossible to test the efficiency of the drug by administering to each patient affected by a particular disease, but we will test it only through a sample. A quality control engineer may be interested in the quality of the products manufactured by a company. He uses a powerful technique called acceptance sampling to protect the producer and consumer interests. An agricultural scientist wanted to test the efficacy of fertilizers should test by designed experiments. He may be interested in farm size, use of land and crop harvested etc. One advantage of working in statistics is that one can combine his interest with almost any field of science, technology or social sciences such as Economics Commerce, Engineering Medicine, and Criminology and so on.

The profession of statistician is exciting, challenging and rewarding. Statistician is the most prevalent title but professionals like Risk analyst, Data analyst, Business analyst have been engaged in work related to statistics. In view of the overwhelming demand for Statistics many universities in India and elsewhere have been offering courses in statistics at graduate and Master's level. We have mentioned earlier that statistics has applications to almost all fields. Here in this section we highlight its applications to select branches.

1.4.1 Statistics and actuarial science

Actuarial science is the discipline that extensively applies statistical methods among other subjects involved in insurance and financial institutions. The professionals who qualify in actuarial science course are called actuaries. Actuaries, in the earlier days used deterministic models to assess the premiums in insurance sector. Nowadays, with modern computers and sophisticated statistical methods, science has developed vastly. In India, from 2006 a statutory body has been looking after the profession of actuaries.

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